Supplemental Material: Prompt Language Doesn't Matter, but Keep the Original Text: Large Language Models and Sentiment Analysis in French News

Abstract

Supplemental tables for the article. Tables S1 to S8 are referred to in the main text.

Keywords: supplemental material

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Table S1
Search Terms Used to Select Articles in the Eureka Database

Group	Search terms
French terms	*logiciel libre*, *logiciels libres*, *logiciel open source*, *logiciels open source*, *code source ouvert*, *code source libre*
English and borrowed terms	*open source*, *open-source*, *free software*, *software libre*
Proper names	*Free Software Foundation*, *Richard Stallman*

Table S2
Collected Articles by Source Media

Source	Articles	First article	Last article
Libération	769	1995-06-23	2024-11-04
Monde, Le	666	2001-01-09	2024-12-26
Figaro, Le	402	1998-10-21	2024-12-04
Devoir, Le	318	1998-07-20	2024-10-26
Soleil, Le (Québec, QC)	179	1998-11-08	2023-11-18
Presse, La	152	1998-03-26	2017-10-14
Journal de Montréal, Le	65	2006-12-20	2023-04-24
Journal de Québec, Le	45	2007-03-24	2024-10-19
Tribune, La (Sherbrooke, QC)	43	2000-03-29	2022-12-03
Droit, Le (Ottawa, ON)	17	1998-11-10	2014-07-17
Nouvelliste, Le (Trois-Rivières, QC)	13	2000-01-13	2022-03-05
Quotidien, Le (Saguenay, QC)	10	2008-11-01	2019-04-10
Voix de l'Est, La (Granby, QC)	4	2010-09-17	2015-02-14

Table S3
Wording of the English Prompt on French Text, by Component

Component	Wording
Task	Please analyze the sentiment of the following French text and provide a single numerical rating according to this scale:
Scale, -1.0	Strong negative sentiment - highly critical, hostile, or pessimistic content
Scale, -0.5	Moderate negative sentiment - somewhat negative, disapproving, or concerned content
Scale, 0.0	Neutral sentiment - factual, balanced, or neither positive nor negative content
Scale, 0.5	Moderate positive sentiment - somewhat positive, approving, or optimistic content
Scale, 1.0	Strong positive sentiment - highly supportive, enthusiastic, or optimistic content
Intermediate values	You may also use values between these points for intermediate sentiment levels (e.g., -0.7, 0.3).
Instruction 1	First, carefully read and understand the text, considering cultural and linguistic nuances in French.
Instruction 2	Analyze the emotional tone, word choice, and overall message.
Instruction 3	Respond ONLY with a single numerical value between -1.0 and 1.0 that best represents the sentiment.
Instruction 4	Do not include ANY explanations, analysis, or additional text in your response.
Input	Here is the text to analyze: [sentence]

Table S4

Correlation With Human Judgments (With 95% Confidence Interval), Mean Absolute Error, and Number of Sentences Scored, by Model and Condition

Model	Weights	Condition	r	95% CI	MAE	n
GPT-5.6 Luna	Closed	FR→FR	0.810	[0.73, 0.89]	0.168	200
GPT-5.6 Luna	Closed	EN→FR	0.802	[0.72, 0.89]	0.165	200
GPT-5.6 Luna	Closed	EN→EN	0.728	[0.63, 0.82]	0.176	200
Gemini 3.5 Flash	Closed	FR→FR	0.770	[0.68, 0.86]	0.158	200
Gemini 3.5 Flash	Closed	EN→FR	0.771	[0.68, 0.86]	0.156	200
Gemini 3.5 Flash	Closed	EN→EN	0.758	[0.67, 0.85]	0.168	200
Jev 1.13	Closed	FR→FR	0.772	[0.68, 0.86]	0.163	200
Jev 1.13	Closed	EN→FR	0.772	[0.68, 0.86]	0.169	200
Jev 1.13	Closed	EN→EN	0.752	[0.66, 0.84]	0.176	200
Claude Haiku 4.5	Closed	FR→FR	0.785	[0.70, 0.87]	0.182	200
Claude Haiku 4.5	Closed	EN→FR	0.780	[0.69, 0.87]	0.185	200
Claude Haiku 4.5	Closed	EN→EN	0.705	[0.61, 0.80]	0.182	200
Qwen3 235B-A22B	Open	FR→FR	0.774	[0.69, 0.86]	0.226	200
Qwen3 235B-A22B	Open	EN→FR	0.753	[0.66, 0.84]	0.232	200
Qwen3 235B-A22B	Open	EN→EN	0.680	[0.58, 0.78]	0.229	200
GPT-OSS 120B	Open	FR→FR	0.727	[0.63, 0.82]	0.205	200
GPT-OSS 120B	Open	EN→FR	0.743	[0.65, 0.84]	0.182	200
GPT-OSS 120B	Open	EN→EN	0.729	[0.63, 0.82]	0.172	200
GPT-OSS 20B	Open	FR→FR	0.733	[0.64, 0.83]	0.171	200
GPT-OSS 20B	Open	EN→FR	0.718	[0.62, 0.82]	0.191	200
GPT-OSS 20B	Open	EN→EN	0.695	[0.60, 0.80]	0.176	200
Mistral Small 3.2	Open	FR→FR	0.735	[0.64, 0.83]	0.263	200
Mistral Small 3.2	Open	EN→FR	0.728	[0.63, 0.82]	0.294	200
Mistral Small 3.2	Open	EN→EN	0.669	[0.57, 0.77]	0.272	200
DeepSeek V3.2	Open	FR→FR	0.686	[0.58, 0.79]	0.203	200
DeepSeek V3.2	Open	EN→FR	0.743	[0.65, 0.84]	0.176	200
DeepSeek V3.2	Open	EN→EN	0.683	[0.58, 0.78]	0.178	200
Mistral Large 4	Closed	FR→FR	0.702	[0.60, 0.80]	0.184	200
Mistral Large 4	Closed	EN→FR	0.713	[0.62, 0.81]	0.176	200
Mistral Large 4	Closed	EN→EN	0.692	[0.59, 0.79]	0.169	200
Qwen3 32B	Open	FR→FR	0.679	[0.58, 0.78]	0.175	200
Qwen3 32B	Open	EN→FR	0.695	[0.59, 0.79]	0.170	200
Qwen3 32B	Open	EN→EN	0.644	[0.54, 0.75]	0.187	200
Llama 4 Scout	Open	FR→FR	0.644	[0.54, 0.75]	0.275	200
Llama 4 Scout	Open	EN→FR	0.661	[0.56, 0.77]	0.286	200
Llama 4 Scout	Open	EN→EN	0.618	[0.51, 0.73]	0.299	200
Llama 3.1 8B	Open	FR→FR	0.612	[0.50, 0.72]	0.273	199
Llama 3.1 8B	Open	EN→FR	0.604	[0.49, 0.71]	0.268	200
Llama 3.1 8B	Open	EN→EN	0.575	[0.46, 0.69]	0.254	200
Lexicoder (EN)	Dictionary	EN	0.548	[0.43, 0.66]	0.203	200
Lexicoder (FR)	Dictionary	FR	0.357	[0.23, 0.49]	0.210	200
Llama 3.2 3B	Open	FR→FR	0.476	[0.35, 0.60]	0.262	200
Llama 3.2 3B	Open	EN→FR	0.132	[-0.01, 0.27]	0.377	200
Llama 3.2 3B	Open	EN→EN	0.302	[0.17, 0.44]	0.305	200
Llama 3.2 1B	Open	FR→FR	0.100	[-0.16, 0.36]	0.551	58
Llama 3.2 1B	Open	EN→FR	0.003	[-0.23, 0.23]	0.604	74
Llama 3.2 1B	Open	EN→EN	-0.199	[-0.44, 0.04]	0.526	66

Table S5

Cost of Scoring 1,000 Sentences at List Prices (Upper Bound; Cached Prompt Tokens Counted at the Full Input Rate; Input Tokens and Cost Not Reported for Llama 3.2 1B, Whose Endpoint Returned a Constant Token Count)

Model	Provider	Weights	API calls per score	Input tokens	Output tokens	USD per 1,000 sentences
GPT-OSS 20B	OpenRouter	Open	1.02	657,496	87,892	0.018
Llama 3.2 3B	OpenRouter	Open	1.00	595,142	7,320	0.018
Jev 1.13	TypeSafe	Closed	1.00	815,199	32,400	0.019
GPT-OSS 120B	OpenRouter	Open	1.00	667,425	95,660	0.020
Qwen3 32B	OpenRouter	Open	1.00	592,086	5,696	0.027
Mistral Small 3.2	OpenRouter	Open	1.00	553,761	7,405	0.029
Qwen3 235B-A22B	OpenRouter	Open	1.00	584,886	7,752	0.030
Llama 4 Scout	OpenRouter	Open	1.00	564,947	8,467	0.059
DeepSeek V3.2	OpenRouter	Open	1.00	559,594	9,340	0.067
Llama 3.1 8B	OpenRouter	Open	1.01	593,398	8,062	0.074
GPT-5.6 Luna	OpenAI	Closed	1.00	552,225	51,819	0.096
Mistral Large 4	OpenRouter	Closed	1.00	534,990	7,603	0.211
Claude Haiku 4.5	Anthropic	Closed	1.00	648,705	12,600	0.395
Gemini 3.5 Flash	Google	Closed	1.03	568,160	658,571	3.766
Llama 3.2 1B	OpenRouter	Open	22.68	–	168,789	–

Table S6
F1 Scores for 3-Category Classification

Model	Negative	Neutral	Positive	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.762	0.375	0.832	0.778
GPT-5.6 Luna (FR→FR)	0.748	0.385	0.800	0.755
Jev 1.13 (EN→FR)	0.691	0.500	0.810	0.751
Jev 1.13 (FR→FR)	0.695	0.400	0.803	0.741
GPT-5.6 Luna (EN→FR)	0.730	0.364	0.776	0.733
Mistral Small 3.2 (FR→FR)	0.578	0.286	0.844	0.721
Jev 1.13 (EN→EN)	0.711	0.222	0.777	0.717
Claude Haiku 4.5 (EN→FR)	0.690	0.273	0.778	0.715
Mistral Small 3.2 (EN→FR)	0.545	0.364	0.841	0.715
DeepSeek V3.2 (FR→FR)	0.672	0.216	0.765	0.697
GPT-5.6 Luna (EN→EN)	0.688	0.367	0.736	0.695
Qwen3 235B-A22B (FR→FR)	0.752	0.243	0.716	0.695
Mistral Small 3.2 (EN→EN)	0.488	0.364	0.836	0.693
DeepSeek V3.2 (EN→FR)	0.696	0.226	0.737	0.688
GPT-OSS 120B (FR→FR)	0.721	0.314	0.712	0.687
Llama 4 Scout (FR→FR)	0.643	0.291	0.747	0.682
Qwen3 235B-A22B (EN→FR)	0.719	0.254	0.706	0.679
Llama 4 Scout (EN→FR)	0.667	0.258	0.732	0.678
GPT-OSS 120B (EN→FR)	0.717	0.324	0.689	0.672
Claude Haiku 4.5 (EN→EN)	0.678	0.256	0.715	0.671
GPT-OSS 20B (EN→FR)	0.700	0.268	0.677	0.655
GPT-OSS 20B (FR→FR)	0.691	0.258	0.680	0.654
Llama 4 Scout (EN→EN)	0.647	0.246	0.686	0.643
Gemini 3.5 Flash (FR→FR)	0.655	0.264	0.673	0.639
DeepSeek V3.2 (EN→EN)	0.672	0.258	0.657	0.633
Mistral Large 4 (EN→EN)	0.604	0.250	0.677	0.624
Llama 3.1 8B (EN→FR)	0.652	0.281	0.646	0.623
Mistral Large 4 (FR→FR)	0.644	0.259	0.650	0.621
Llama 3.2 3B (FR→FR)	0.465	0.195	0.747	0.620
Llama 3.1 8B (EN→EN)	0.627	0.265	0.653	0.618
Gemini 3.5 Flash (EN→FR)	0.642	0.242	0.646	0.616
Mistral Large 4 (EN→FR)	0.648	0.235	0.632	0.609
Llama 3.1 8B (FR→FR)	0.615	0.174	0.643	0.604
Gemini 3.5 Flash (EN→EN)	0.667	0.233	0.613	0.603
GPT-OSS 20B (EN→EN)	0.614	0.234	0.635	0.601
Qwen3 235B-A22B (EN→EN)	0.698	0.227	0.591	0.600
Lexicoder (EN)	0.574	0.231	0.636	0.588
GPT-OSS 120B (EN→EN)	0.678	0.218	0.552	0.569
Qwen3 32B (EN→FR)	0.554	0.165	0.621	0.568
Lexicoder (FR)	0.474	0.289	0.631	0.557
Qwen3 32B (FR→FR)	0.515	0.177	0.611	0.550
Llama 3.2 3B (EN→EN)	0.167	0.261	0.730	0.520
Qwen3 32B (EN→EN)	0.559	0.190	0.520	0.509
Llama 3.2 3B (EN→FR)	0.000	0.222	0.757	0.481
Llama 3.2 1B (FR→FR)	0.000	0.138	0.648	0.443
Llama 3.2 1B (EN→FR)	0.000	0.000	0.568	0.338

Table S7

F1 Scores for 7-Category Classification

Model	Very neg.	Negative	Somewhat neg.	Neutral	Somewhat pos.	Positive	Very pos.	Weighted F1
Claude Haiku 4.5 (FR→FR)	0.400	0.680	0.455	0.375	0.540	0.612	0.000	0.537
Jev 1.13 (EN→FR)	0.200	0.586	0.282	0.500	0.623	0.548	0.250	0.528
Jev 1.13 (FR→FR)	0.222	0.621	0.324	0.400	0.601	0.540	0.250	0.523
Jev 1.13 (EN→EN)	0.200	0.525	0.385	0.222	0.610	0.566	0.286	0.518
GPT-5.6 Luna (FR→FR)	0.286	0.500	0.393	0.385	0.538	0.462	0.133	0.479
GPT-5.6 Luna (EN→EN)	0.364	0.500	0.338	0.367	0.510	0.542	0.250	0.470
GPT-5.6 Luna (EN→FR)	0.333	0.500	0.258	0.364	0.564	0.407	0.125	0.455
DeepSeek V3.2 (EN→EN)	0.333	0.533	0.419	0.258	0.415	0.588	0.667	0.448
Claude Haiku 4.5 (EN→EN)	0.286	0.642	0.400	0.256	0.409	0.568	0.000	0.448
GPT-OSS 120B (EN→FR)	0.333	0.566	0.400	0.324	0.416	0.485	0.133	0.434
Mistral Large 4 (EN→EN)	0.333	0.651	0.316	0.250	0.444	0.408	0.000	0.423
DeepSeek V3.2 (EN→FR)	0.444	0.667	0.429	0.226	0.349	0.493	0.000	0.416
Mistral Small 3.2 (EN→EN)	0.333	0.345	0.222	0.364	0.549	0.364	0.091	0.416
Mistral Large 4 (EN→FR)	0.167	0.600	0.357	0.235	0.361	0.590	0.286	0.416
DeepSeek V3.2 (FR→FR)	0.333	0.368	0.469	0.216	0.400	0.527	0.167	0.415
Claude Haiku 4.5 (EN→FR)	0.500	0.680	0.364	0.273	0.347	0.511	0.200	0.413
Mistral Large 4 (FR→FR)	0.154	0.488	0.344	0.259	0.397	0.561	0.222	0.412
Mistral Small 3.2 (FR→FR)	0.500	0.400	0.170	0.286	0.509	0.380	0.125	0.394
GPT-OSS 20B (EN→FR)	0.235	0.431	0.269	0.268	0.370	0.535	0.000	0.376
GPT-OSS 120B (EN→EN)	0.444	0.692	0.281	0.218	0.288	0.536	0.286	0.373
Gemini 3.5 Flash (FR→FR)	0.222	0.583	0.250	0.264	0.310	0.571	0.200	0.370
GPT-OSS 120B (FR→FR)	0.308	0.545	0.333	0.314	0.319	0.447	0.154	0.369
GPT-OSS 20B (FR→FR)	0.364	0.600	0.286	0.258	0.325	0.448	0.286	0.367
GPT-OSS 20B (EN→EN)	0.400	0.528	0.235	0.234	0.338	0.508	0.000	0.362
Gemini 3.5 Flash (EN→FR)	0.400	0.694	0.255	0.242	0.263	0.485	0.167	0.349
Qwen3 32B (EN→FR)	0.222	0.512	0.163	0.165	0.286	0.615	0.000	0.334
Qwen3 235B-A22B (FR→FR)	0.190	0.407	0.320	0.243	0.319	0.369	0.087	0.330
Mistral Small 3.2 (EN→FR)	0.364	0.312	0.133	0.364	0.420	0.292	0.062	0.326
Gemini 3.5 Flash (EN→EN)	0.364	0.652	0.222	0.233	0.230	0.500	0.000	0.324
Qwen3 32B (FR→FR)	0.333	0.558	0.125	0.177	0.236	0.632	0.000	0.315
Qwen3 235B-A22B (EN→FR)	0.182	0.321	0.240	0.254	0.350	0.339	0.091	0.315
Qwen3 235B-A22B (EN→EN)	0.182	0.275	0.302	0.227	0.279	0.408	0.133	0.299
Llama 3.1 8B (EN→FR)	0.111	0.120	0.192	0.281	0.319	0.507	0.200	0.297
Lexicoder (FR)	0.000	0.000	0.253	0.289	0.479	0.057	0.000	0.290
Llama 4 Scout (FR→FR)	0.125	0.167	0.182	0.291	0.341	0.356	0.100	0.286
Lexicoder (EN)	0.000	0.000	0.267	0.231	0.472	0.000	0.000	0.276
Llama 3.1 8B (FR→FR)	0.143	0.197	0.179	0.174	0.239	0.543	0.000	0.267
Llama 3.2 3B (FR→FR)	0.000	0.364	0.200	0.195	0.240	0.395	0.000	0.266
Qwen3 32B (EN→EN)	0.222	0.364	0.172	0.190	0.168	0.567	0.333	0.261
Llama 3.1 8B (EN→EN)	0.138	0.222	0.136	0.265	0.246	0.479	0.143	0.261
Llama 4 Scout (EN→FR)	0.154	0.261	0.143	0.258	0.281	0.329	0.080	0.257
Llama 4 Scout (EN→EN)	0.105	0.280	0.125	0.246	0.234	0.434	0.154	0.252
Llama 3.2 3B (EN→EN)	0.000	0.138	0.050	0.261	0.250	0.375	0.000	0.217
Llama 3.2 3B (EN→FR)	0.000	0.000	0.000	0.222	0.232	0.317	0.000	0.171
Llama 3.2 1B (EN→EN)	0.000	0.000	0.000	0.150	0.000	0.211	0.000	0.051
Llama 3.2 1B (FR→FR)	0.000	0.000	0.000	0.138	0.000	0.000	0.000	0.007
Llama 3.2 1B (EN→FR)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

Table S8

Mean Input Tokens per API Call by Language Condition, and Tokens per Word by Text Language (Valid Calls Only; Llama 3.2 1B Excluded Because Its Endpoint Reported a Constant Token Count on Every Call; Jev's Input Is a Structured Question Rather Than the Text Prompt, So Its Counts Are Not Comparable)

Model	EN→EN	EN→FR	FR→FR	French-prompt overhead	Tokens per French word	Tokens per English word
Mistral Large 4	284.5	289.1	318.1	+29 (+10.0%)	1.38	1.22
Jev 1.13	431.0	440.3	487.3	+47 (+10.7%)	1.62	1.22
GPT-OSS 120B	352.0	357.7	402.7	+45 (+12.6%)	1.42	1.19
GPT-OSS 20B	339.0	344.7	389.7	+45 (+13.1%)	1.42	1.19
Llama 4 Scout	294.1	299.9	343.9	+44 (+14.7%)	1.41	1.19
Mistral Small 3.2	288.6	294.6	339.6	+45 (+15.3%)	1.45	1.22
GPT-5.6 Luna	288.0	293.7	338.7	+45 (+15.3%)	1.42	1.19
Gemini 3.5 Flash	285.0	292.1	337.1	+45 (+15.4%)	1.46	1.17
Qwen3 32B	296.5	307.7	382.7	+75 (+24.4%)	1.69	1.21
DeepSeek V3.2	280.1	290.3	361.3	+71 (+24.5%)	1.61	1.19
Qwen3 235B-A22B	292.5	303.7	378.7	+75 (+24.7%)	1.69	1.21
Llama 3.2 3B	295.2	306.8	388.8	+82 (+26.7%)	1.69	1.20
Llama 3.1 8B	295.2	306.8	389.0	+82 (+26.8%)	1.69	1.20
Claude Haiku 4.5	313.8	328.2	439.2	+111 (+33.8%)	1.87	1.25